

Short Questions 6

03/01/2026

Your answers should be formal with mathematical formulas and algorithms, as necessary. Type your responses. Upload a PDF file. If you want to add a drawing, you can include a nicely hand-drawn figure or a figure drawn with a computer tool. Use single spaced typing. Preferably, keep your answers to a page.

1. (2 points) What are some similarities and differences between Monte Carlo RL and Temporal Difference RL?
2. (3 points) What is the update formula for the famous table-based algorithm called Q-learning? What is the “philosophy” behind how the agent acts and learns in Q-learning?
3. (5 points) To overcome the main shortcoming of Q-learning, researchers have proposed many different approaches. One approach uses two estimators of q values, instead of one. What is the purpose of using two tables instead of one in this modified form of Q-learning? Write down the formulas for updating table entries.